

OTTAWA NATIONAL FOREST SILVICULTURAL PRESCRIPTION / MARKING GUIDES

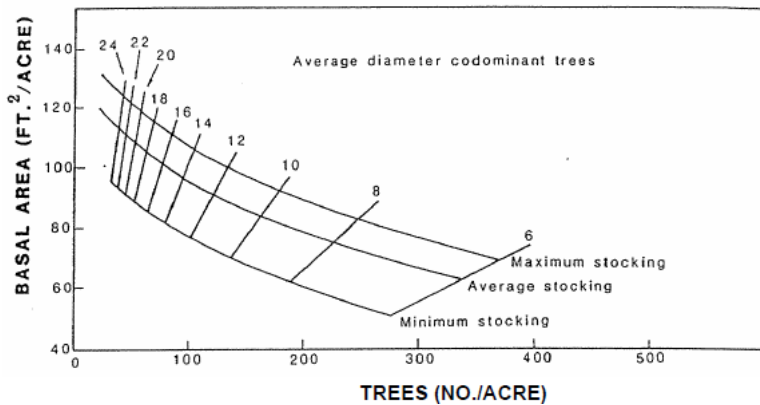
NEPA DECISION DOCUMENT: Grosbeck	TIMBER SALE: Jackalope	PU: 1 - 9
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STAND INFORMATION												
Ranger District	Comp-Stand	Acres	MA	LTA	ELTP	Habitat Type	Cover Type	LSC	Year of Origin	Size Density	SI	BA
2	161-3		2.1	11	420B	ATD	89	500		6	51	140
2	161-2		2.1	11	423A	TMC	20	500	1880	9	47	140
2	161-13		2.1	11	420B	ATD	89	500		6	52	130
2	161-15		2.1	11	423C	TMC	89	500	1918	6	48	130
2	160-16		2.1	11	423C	TMC	20	500	1922	6	55	150
2	160-18		2.1	11	427B	ATD	21	500		6	54	200
2	160-20		2.1	11	427B	ATD	89	500		6	45	130
2	160-24		2.1	11	427B	ATD	20	500		9	51	120
2	160-37		2.1	11	427B	ATD	81	500		9	54	130
2	160-36		2.1	11	427B	ATD	89	500		9	49	140
2	160-22		2.1	11	426B	AVO	89	500		9	47	130

STAND DESCRIPTION

Average BA of the stands is between 120 – 200 Ft²/Ac. Stand 18 has the highest BA. Average DBH ranges from 10 – 16 inches. The majority of these stand have a component of hemlock mostly in the overstory however stand 16 has a fair amount of hemlock seedlings around knee high. Maple is the dominant species with hemlock, yellow birch, basswood and cherry in lesser amounts. There are pockets of maple regeneration mostly in the seedling size. Quality in these stands range from marginal to poor.

– Average Stand stocking



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DESIRED CONDITION

Healthy uneven aged northern hardwood stands with improvement in tree quality, maintaining or increasing current species diversity, maintaining current diameter distribution and establishing a new cohort. Continuing to build on quality and species diversity.

SHORT AND LONG TERM OBJECTIVES

- Apply silvicultural treatments to regenerate and restore a healthy uneven-aged northern hardwood stands.
- Target a residual basal area of 70 - 80 FT²/AC taking no more than a third of the initial TPA.
- Retain or increase species diversity by retaining species other than sugar maple.
- Retain and release healthy white pine, hemlock, cedar, oak, and elm.
- Achieve a stocking level of 3,000 seedlings per acre of native species.
- Maintain and protect ecological processes and functions in and around riparian areas where they occur within the sale area.

REMARKS

There are scattered, small, poorly drained pockets, seeps, steep slopes, perched water tables, or shallow bedrock areas; do not designate any trees for removal in these areas. All spruce and balsam should be considered high risk and removed.

100% of conifer volume should be considered salvage due to spruce decline. Aspen salvage volume should follow collection rate letter dated 1/27/2025. 15% of hardwood volume should be considered salvage due to insect/disease and overstocked conditions.

MARKING GUIDE:

Name of Treatment - Single Tree Selection

Residual BA/Ac – 70 – 80
FT²/Ac (remove no more
than a third of initial BA)

Acres - See Marking Map

Target a residual basal area of 80 FT²/AC. Don't mark below 70 FT²/Ac to reduce the spread of sedge. In areas with high initial BA (> 130 FT²/Ac) generally don't remove more than 30% of the initial BA, however you need to ensure operability. PU 6 will have a higher residual BA due to a starting BA of 200 FT²/Ac. Stand should have a diameter distribution that mirrors the parent stand. Remove high risk trees that are diseased/structurally damaged unless it's a cull or provides wildlife use. Consider all ironwood, spruce and balsam high risk and remove all. Retain and promote healthy yellow & paper birch, hemlock, basswood, oak, white pine, cedar, and cherry to maintain or increase diversity. Some of these species, yellow birch, basswood & black cherry, may need to be removed due to poor form, quality, health or maturity. Remove the majority of black cherry with black knot and black cherry that are over 16".

Order of Removal For All Payment Units:

1. Target high risk trees that are diseased or will most likely die within the next 15 years that still contain merchantable products. Release hemlock and yellow birch on all 4 sides. Release advanced hemlock regeneration.
2. Promote quality development though removing competing trees of lesser quality to release the best crop trees. Competition among closely spaced crop trees can be retained to preserve the quality of the stand while still maintaining the target residual BA. In areas lacking wildlife trees or stags, trees lacking merchantable products can be retained for future wildlife snag/cavity trees. Don't be afraid to remove poor form/quality yellow birch while retaining the best form for seed trees.
3. If still above BA objective: continue to remove unacceptable growing stock by targeting all size classes that exhibit signs of structural/growth/disease that may be inhibiting the current or future growth potential while improving the growing space of the residual stems as identified in the table below. Choose to remove maple species first to increase species diversity.

DESIGN CRITERIA AND MITIGATION MEASURES

- Cull trees should be left for wildlife, do not remove trees with little or no value unless they pose a safety hazard. If cull trees need to be marked the silviculturist should be notified to look at the risk.
- If any stick nests are found the wildlife biologist should be consulted.
- Reserve snags and den trees according to Forest Plan guidelines (p. 2-32); focus on retaining trees with features beneficial to the NLEB such as shaggy bark and cavities.
- If any archaeological sites are located during layout or implementation the Archaeologist should be consulted.

Within all harvest units except aspen clearcuts, favor hemlock, white pine, cedar, elm, and oak by retaining and crown releasing these species as directed in the stands' silviculture prescription. These species may be cut to facilitate harvest operations where necessary, or to improve growing space and vigor among these same species that may occur within inclusions. The objective for retaining these species is to improve and/or maintain structural and compositional diversity (Forest Plan, p. 2-2).

POST HARVEST TREATMENTS

FACTS Code	Name of Treatment	Prescription
4341	1 st Year Stocking Survey	1 st year stocking survey
4341	3 rd Year Stocking Survey	3 rd year stocking survey
4382	Certification for Naturals wo/SP	Enter certification once regeneration goal is met.

ACTIVITIES SUMMARY

Year	Facts Activity Code	Description
2030	4151	Single-Tree Selection Harvest
2031	4341	1 st year stocking survey
2033	4341	3 rd year stocking survey
2033	4382	Certification for Naturals wo/SP

Prepared by: Michael Strieter	Title: Forest Silviculturist	Date: 2/24/2026
Reviewed by: Rachel Tarpey	Title: TMA	Date: 02/26/2026
Reviewed by:	Title:	Date:
Approved by: Michael Strieter	Title: Certified Silviculturist	Date: 2/24/2026

JACKALOPE T.S. – ESTIMATED ACRES
CRUISE DESIGN & MARKING INSTRUCTIONS

Sale Number: 21035

Estimated Sale Value: \$XXXXXX

Estimated CV Ranges: 15-45%

Maximum Sampling Error: 10%

Based on Cruise Data: Pre-Cruise plots

Cruise Methods: Sample Tree (STR)

[Reference attached Minimum Tree and Piece Utilization Standards, dated 11/02/2015]

[Total Sale Acres = 535 Acres including 1 acre ROW)]

Jackalope Payment Unit & Acreage Summary Table

P.U. #	Estimated Acres	Final Acres	Strata	Marking Method	Species
P.U. 01	51	-	1	CTM	All Species
P.U. 02	58	-	1	CTM	All Species
P.U. 03	36	-	1	CTM	All Species
P.U. 04	40	-	1	CTM	All Species
P.U. 05	75	-	1	CTM	All Species
P.U. 06	70	-	1	CTM	All Species
P.U. 07	81	-	1	CTM	All Species
P.U. 08	67	-	1	CTM	All Species
P.U. 09	56	-	1	CTM	All Species
P.U. 10 (R.O.W.)	1	1	1	CTM	All Species

JACKALOPE T.S. – ESTIMATED ACRES **CRUISE DESIGN & MARKING INSTRUCTIONS**

Single-Tree Selection Payment Units: 01-09 (534 Acres).

Payment Unit 10 (R.O.W.) (1 acre) will be marked and cruised by USFS.

Sample Group	Species/Product (Unit)	DBH Range	Sample Interval
HS	Small Hardwood/Pulpwood (02)	5.0"-10.99"	1:350
HL	Large Hardwood/Pulpwood (02)	11.0"+	1:50
SS	Small Sugar Maple/Sawtimber (01)	11.0"-14.99"	1:30
SL	Large Sugar Maple/Sawtimber (01)	15.0"+	1:15
OS	Small Other Hardwood/Sawtimber (01) <i>[includes Red Maple, Basswood, Black Cherry, Yellow Birch, White Birch, & Red Oak]</i>	11.0"-14.99"	1:30
OL	Large Other Hardwood/Sawtimber (01) <i>[includes Red Maple, Basswood, Black Cherry, Yellow Birch, White Birch, & Red Oak]</i>	15.0"+	1:15
FP	Ash/Pulpwood (02) <i>[includes Black Ash & White Ash]</i>	5.0"+	1:30
FW	Ash/Sawtimber (01) <i>[includes Black Ash & White Ash]</i>	11.0"-15.0"+	1:10
AS	Small Aspen/Pulpwood (02)	5.0"-14.99"	1:10
AL	Large Aspen/Pulpwood (02)	15.0"+	1:10
CS	Small Mixed Conifer/Pulpwood (02) <i>[includes Balsam Fir, Tamarack, Cedar, Hemlock, White Pine, Scots Pine, White Spruce, Black Spruce, & Norway Spruce]</i>	5.0"-14.99"	1:50
CL	Large Mixed Conifer/Pulpwood (02) <i>[includes Balsam Fir, Tamarack, Cedar, Hemlock, Scots Pine, White Spruce, Black Spruce, & Norway Spruce]</i>	15.0"+	1:20
PP	Pine/Pulpwood (02) <i>[includes Red Pine & Jack Pine]</i>	5.0"+	1:1
PW	Pine/Sawtimber (01) <i>[includes Red Pine & White Pine]</i>	9.0"+	1:1

Note: Cruise all Aspen, Balsam Fir, Spruce, & Jack Pine as pulpwood.

Marking Instructions

JACKALOPE T.S. – ESTIMATED ACRES **CRUISE DESIGN & MARKING INSTRUCTIONS**

Hardwood Selection/Sample Tree Cruising (STR): Payment Units: 01-09 (534 Acres).

ALL merchantable trees to be removed from this Payment Unit require two paint marks at DBH on opposite sides of the tree *AND* a corresponding stump mark (see specific Marking Instructions for this Payment Unit). Cruise marked trees using the Sample Groups and Sample Intervals above.

A random start should be used for each sample group at the start of marking in each payment unit. Select a random interval that is less than the required sample interval for the first tally in each sample group, thereafter, sample at the specified sample interval.

Maintain an accurate tree count by payment unit for each sample group. Original field records used for tracking tree counts must be with the tally cards or data recorder sheets when each unit is completed. Do not recopy field notes of tree counts or tally cards.

Right-of-Way Clearing (USFS mark)/Sample Tree Cruising (STR): Payment Unit 10 (1 acre)

ALL merchantable trees to be removed from this Payment Unit require paint marks at DBH on opposite sides of the tree *AND* a corresponding stump mark. Cruise marked trees using the Sample Groups and Sample Intervals above.

NOTE: Refer to Marking Guides for specific marking instructions for these Payment Units.

** USFS personnel will be marking Payment Units 10- road R.O.W. at a later date.*

NOTE: Government furnished tree marking paint (TMP) will be issued or returned by appointment only, at the WATERSMEET RANGER DISTRICT OFFICE. The primary point of contact for all TMP transactions will be the paint custodian. This person will be identified at the pre-work meeting. BLUE paint has been designated for this sale.

JACKALOPE T.S. – ESTIMATED ACRES
CRUISE DESIGN & MARKING INSTRUCTIONS

Cruise Design Prepared by: Hayley Jardeen, Forestry Technician Date: 02/23/2026
Final Acres Updated by: Hayley Jardeen, Forestry Technician Date: 02/23/2026

Cruise Design Reviewed by: Brandon Pertile, Silviculture Technician Date: _____
Cruise Design Reviewed & Updated by: _____ Date: _____

Marking Contract Narrative by: Hayley Jardeen Date: 02/23/2026

Cruise Design & Marking Instructions Approved by: Rachel Tarpey Date: 02/26/2026

JACKALOPE T.S. – ESTIMATED ACRES **CRUISE DESIGN & MARKING INSTRUCTIONS**

MINIMUM TREE AND PIECE UTILIZATION STANDARDS (11/02/2015 revision)

		Minimum Tree Specifications	Minimum Piece Specifications		
SPECIES	PRODUCT	DBH (inches)	Length (feet)	DIB (inches)	Minimum Net Scale in % of Gross
Hardwoods (except aspen)	Sawlogs	11.0	8	9.6-11.5	95% ^{1/}
Hardwoods	Sawlogs	11.0	8	11.6+	50% ^{1/}
Softwood	Sawlogs	9.0	8	7.6	40
All Species	Pulpwood	5.0	8	4.0	70 and reasonably straight ^{2/}

^{1/} In Timber Sale Contracts, all hardwood sawlogs will use 50% minimum net scale in gross %, due to limitations in TIM. Cruising will use 95% minimum net scale for small diameter logs (9.6 – 11.5 DIB).

^{2/} Seventy percent applies to rot, voids, and char. Mechanical type defects such as sweep, crook, spider heart, and ring shake will not be considered.

Reasonably straight - When the true line of a minimum length piece does not deviate more than ½ the inside diameter of the small and plus two inches from a straight line drawn between the centers of the ends of a piece.

The maximum deflection allowed is what will fit in a 20-inch cylinder.

A minimum tree must include at least one piece that meets minimum specifications.

**** Product heights are now measured to the nearest foot. All heights are measured in feet from the ground on the high side of the stump.**

- 1. Conifers – measure total height (to the tip) in feet and the cruise program will calculate the 4” cut-off.**
- 2. Hardwoods (includes Aspen) – measure the height in feet to the 4” top.**
- 3. Sawtimber – cut-off point or “stoppers” will follow our regular practices. Measure sawtimber height in feet. Secondary product height will be measured as described in item 1 or 2 depending on the species.**
- 4. Default dib’s are already built into the equations;**
 pulpwood = 4.0”
 sawtimber = 9.6” (hardwood)
 = 7.6” (mixed conifers, pine)
- 5. On trees with a broken, dead, or deformed top, record the height to the point where the 4” top or the tip would have been if the tree were still growing normally, then record defect to account for the missing volume.**